

UniEval-RAG: A Unified End-to-End Evaluation Framework for RAG Systems

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Abstract—Retrieval-Augmented Generation (RAG) reduces hallucinations commonly observed in large language models (LLMs) by integrating external document retrieval. However, existing research lacks a rigorous evaluation of the entire RAG pipeline. Current evaluation frameworks focus on the retrieval and generation stages while neglecting systematic metrics for upstream processes such as text chunking. Furthermore, they have yet to establish a unified evaluation system covering all stages of the RAG pipeline. To address this gap, we introduce UniEval-RAG, a unified evaluation framework that provides end-to-end coverage of the entire RAG pipeline. In addition to evaluating the retrieval and generation stages, this framework incorporates novel metrics, such as keyword density and keyword coverage, to assess the structural coherence and semantic completeness of chunking. Additionally, we introduce a benchmark dataset covering five question answering (QA) scenarios—factual, comparative, analytical, tutorial, and summarization—to support diverse evaluation needs. Experimental results demonstrate that UniEval-RAG effectively identifies performance bottlenecks and offers interpretable insights to guide the optimization of RAG systems.

Index Terms—Nature Language Process (NLP); Retrieval-Augmented Generation (RAG); Unified Evaluation Framework; Chunking Metrics; Keyword Density and Coverage; Multi-Granularity Keywords; Question Answering Evaluation

I. INTRODUCTION

Retrieval-Augmented Generation (RAG) [1] combines retrieval models with large language models (LLMs) [2]–[5], enabling the latter to leverage external knowledge sources for generating answers. This integration has been shown to help mitigate the hallucination problem commonly observed in LLMs [6]–[8]. Consequently, RAG has been widely adopted due to its demonstrated effectiveness in knowledge-intensive tasks. However, with the rapid advancement of RAG systems, developing comprehensive and fine-grained evaluation methods has emerged as a critical challenge in current research.

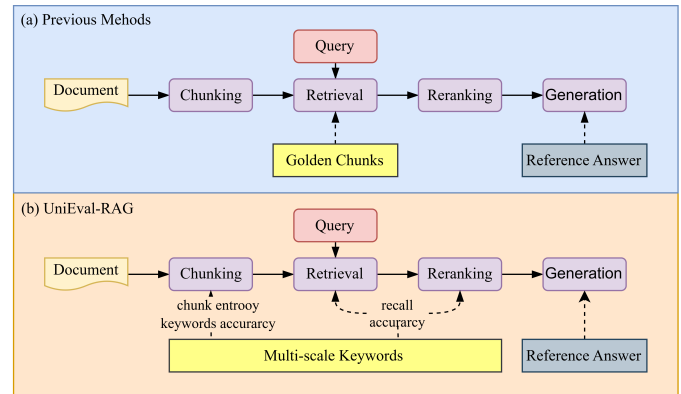


Fig. 1: Overview of previous methods and the proposed UniEval-RAG for evaluating RAG systems.

On one hand, RAG systems typically consist of multiple modules—chunking, retrieval, re-ranking, and generation—each with distinct functional objectives [9]. Therefore, the evaluation metrics and methods should be tailored to the specific roles of these modules. On the other hand, the final output quality of RAG systems is often susceptible to error propagation across these stages. For instance, if the chunking stage omits information, the resulting answer may be incorrect. Similarly, if the retrieval module returns semantically irrelevant documents, even powerful LLMs will struggle to generate accurate responses [10]. Therefore, quantifying the contribution of each stage to overall system performance and developing targeted evaluation strategies are essential for improving the reliability and real-world applicability of RAG systems.

Although several evaluation frameworks have explored the performance of RAG systems, current mainstream approaches

still suffer from three major limitations:

(1) **Over-reliance on final-answer metrics.** Most evaluation methods remain heavily reliant on traditional QA datasets, focusing primarily on the accuracy of the final generated answers [11]–[14]. However, they often overlook the independent impact of intermediate stages. This evaluation approach makes it difficult to identify which specific stage constitutes the performance bottleneck, thereby hindering transparent and interpretable diagnosis throughout the system pipeline.

(2) **Neglect of explicit chunking evaluation.** Existing evaluation techniques generally lack explicit assessment of the chunking stage [11]. As the initial step in the RAG pipeline, chunking directly affects both the completeness of information retention and the contextual relevance for subsequent retrieval. Nevertheless, current methods often infer chunking quality indirectly through observed changes in QA performance or retrieval effectiveness. Such implicit evaluations lack interpretability and offer limited actionable guidance for system optimization. Therefore, developing quantifiable and interpretable metrics for chunking quality is a necessary prerequisite for the fine-grained optimization of RAG systems.

(3) **Lack of end-to-end evaluation frameworks.** Although some studies conduct stage-level analyses of RAG systems, they often treat each stage as an isolated module, primarily focusing on comparing different embedding models or LLMs [15]. A comprehensive evaluation of the entire RAG pipeline remains lacking. This gap hinders the meaningful comparison of overall system performance and prevents the accurate quantification of how improvements in upstream modules affect end-to-end outcomes.

To address the aforementioned challenges, we introduce a unified end-to-end evaluation framework for RAG systems, known as UniEval-RAG. This framework systematically covers the four critical stages of RAG: chunking, retrieval, re-ranking, and generation, enabling horizontal comparisons across various RAG systems and precise identification of system bottlenecks. Notably, UniEval-RAG innovatively introduces explicit evaluation metrics for chunking quality, incorporating keyword density and keyword precision as measures to assess chunking rationality from both structural coherence and semantic coverage perspectives. Furthermore, we construct a comprehensive dataset spanning five QA scenarios—factual, analytical, comparative, tutorial, and summarization—to enrich the diversity and challenge of evaluation tasks. In summary, the main contributions of this work include:

- The first explicit metric for evaluating chunking quality was introduced, incorporating keyword density and keyword precision to measure structural rationality and semantic coverage at the chunking stage.
- The development of UniEval-RAG, a unified end-to-end evaluation framework covering chunking, retrieval, re-ranking, and generation stages, addresses the inability of traditional evaluation methods to compare performance differences across RAG system components.
- The design of a more comprehensive dataset construction methodology incorporates five diverse QA types, enhanc-

ing the breadth and representativeness of the evaluation.

II. RELATED WORKS

A. Retrieval-Augmented Generation

In recent years, a substantial body of research on Retrieval-Augmented Generation (RAG) has emerged both domestically and internationally [1], [16]–[22]. Lewis et al. [1] were the first to propose the RAG model, which integrates large pre-trained generative models with explicit retrieval mechanisms to enhance performance on knowledge-intensive tasks. Subsequent work has focused on improving retrieval model accuracy and designing more efficient generation prompting strategies. For example, Zhao et al. [21] introduced novel chunking strategies and evaluation metrics specifically for Chinese RAG systems to improve retrieval-based question answering performance; Lyu et al. [22] proposed the CRUD-RAG benchmark to assess RAG performance across different scenarios comprehensively—create, read, update, and delete. These efforts have advanced RAG technology, but primarily concentrate on the models themselves or specific application contexts, with relatively little focus on unified evaluation across the entire RAG pipeline.

B. Retrieval-Augmented Generation Evaluation

Regarding RAG evaluation frameworks, several attempts have been made [11], [22]–[25]. Es et al. [24] introduced RAGAs (Retrieval Augmented Generation Assessment), which employs a series of automated metrics to evaluate the relevance of retrieved contexts and the quality of generated answers without requiring human references. Saad-Falcon et al. [11] proposed the ARES framework, which utilizes generative language models as evaluators; it trains assessors on synthetic data to automatically score dimensions such as context relevance, answer faithfulness, and correctness. Additionally, domestically, Lyu et al. [22] extended the question answering benchmark in CRUD-RAG with diverse task types to examine the component-level effects within RAG systems. Overall, these frameworks have significantly enriched the RAG evaluation toolbox; however, they tend to focus either on retrieval outputs (e.g., RAGAs emphasizing context relevance) or on generation results (e.g., ARES focusing on answer accuracy), lacking integrated evaluation across all pipeline stages.

III. PRELIMINARIES

RAG typically comprises four key stages: chunking, retrieval, re-ranking, and answer generation. Chunking serves as the initial step in the pipeline, involving the segmentation of raw corpora into smaller, structured information units to facilitate finer-grained indexing and matching by subsequent retrieval modules [26]–[28]. The retrieval stage is responsible for selecting a set of text passages or document fragments from an external knowledge base that are most relevant to the user query, often achieved through vector similarity computations or matching mechanisms based on sparse or dense indices [29]–[31]. The re-ranking stage further refines the preliminary

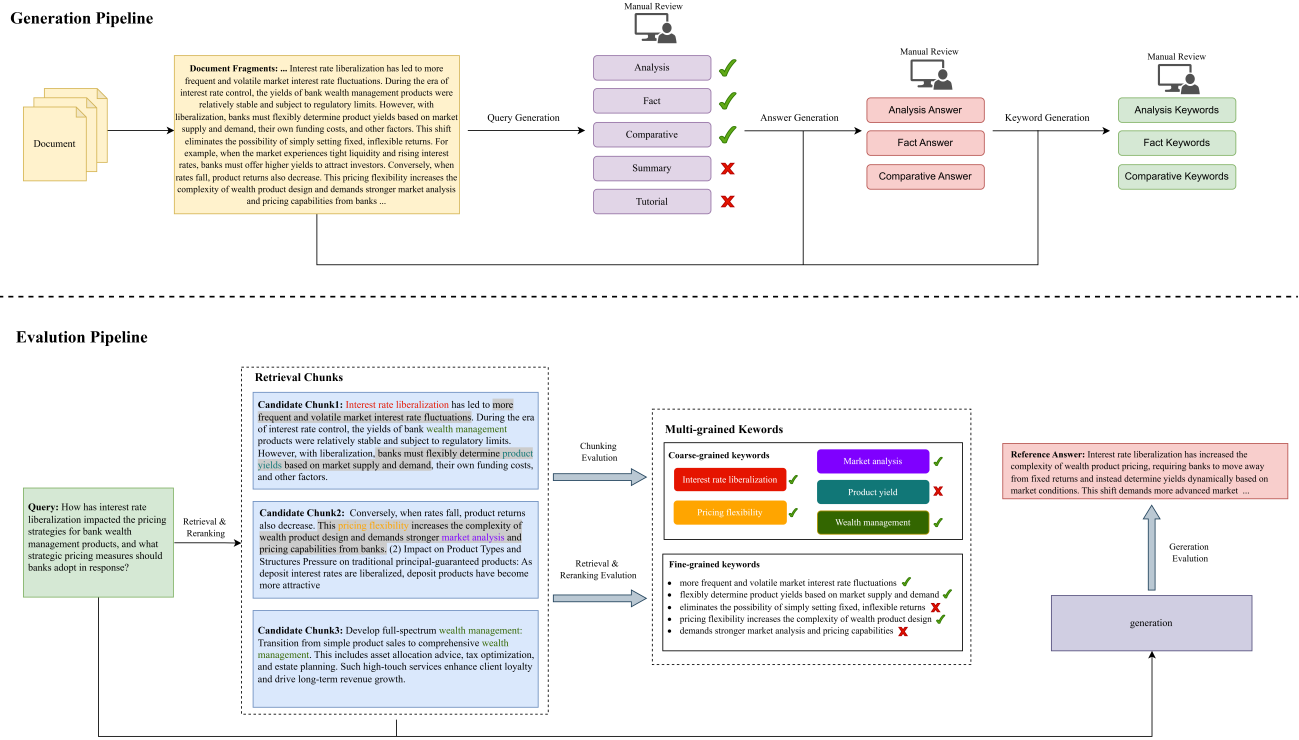


Fig. 2: The overall framework of UniEval-RAG. The generation pipeline refers to the process of keyword generation, while the evaluation pipeline denotes the procedure through which UniEval-RAG performs evaluation.

retrieval results by performing fine-grained semantic reordering to enhance the semantic relevance of retrieved items. [22], [32], [33] Finally, the answer generation stage synthesizes the input query and retrieved evidence to produce natural language answers using large language models. for you.

IV. THE UNIEVAL-RAG FRAMEWORK

In this section, we provide a detailed introduction to the proposed UniEval-RAG framework, which aims to evaluate all stages of RAG systems, including chunking, retrieval, re-ranking, and answer generation. We introduce multi-granularity keywords to facilitate the assessment of chunking, retrieval, and re-ranking performance. Figure 2 illustrates the detailed process of the proposed UniEval-RAG framework.

A. Data Collection

We collect documents in various formats, including DOC, PDF, and XLSX, from multiple open-source platforms. These documents cover diverse domains such as finance, technology, humanities and social sciences, and healthcare. The time span of the documents extends from 2025 to the present, which is notably later than the training data cutoff for Deepseek-v3 [34], [35], ensuring the novelty and timeliness of the evaluation data. The document contents include plain text, structured data tables, and mixed data formats. The proportion of each document type within the overall dataset is summarized in Table I.

TABLE I: Distributions of different file types

File Type	Avg. Tokens	Avg. Pages	Count
PDF	25,770.8	8.4	67
DOC	12,627.5	11.8	53
XLSX	3,245.3	2.6	34

B. Query Generation

We design five types of questions to cover common information needs: factual, comparative, analytical, tutorial, and summarization. Definitions and examples for each question type are provided in Table II. Based on these, we combined document segments with carefully crafted prompt templates to automatically generate corresponding queries using Deepseek-v3.

The query generation process strictly follows the guidelines proposed by CoFE-RAG [9], which include:

- Queries must be semantically explicit, avoiding ambiguous referential terms such as “it” or “this.”
- Queries must conform to the definitions of their respective question types.
- Queries should be logically derivable from the corresponding document chunks to ensure traceability.

To ensure quality, experienced human annotators conduct acceptability screening of the generated queries. The acceptability evaluation involve scoring the relevance between queries and document segments on a scale from 1 to 5, with

scores above 4 considered acceptable. The final acceptance rate at this stage is 83.2%.

TABLE II: Definitions of five types of queries.

Type	Description
Analytical	Analytical explanation or synthesis of specific concepts, terms, or phenomena.
Comparative	Comparative examination across multiple dimensions or criteria.
Factual	Verifiable facts or empirical evidence with concrete specificity.
Tutorial	Stepwise procedural guidance for task completion.
Summary	Condensed synthesis of complex information or multi-perspective content.

C. Multi-Scale Keyword Generation

To address the limitations of traditional evaluation methods—such as strong dependence on standard chunks and limited annotation flexibility—the multi-scale keyword generation strategy is used to annotate the core semantic content of queries and document segments. This strategy abandons reliance on standard chunk annotations, thereby reducing manual annotation effort and enhancing flexibility.

We introduce two scales of keywords:

- **Coarse-grained keywords:** Broadly representative, capable of covering the core topics of both queries and document segments;
- **Fine-grained keywords:** Focused on localized content within document segments, typically used for answer-level reasoning or detailed questions.

The keyword generation process is as follows: we concatenate queries and document segments and then pass them to Deepseek-v3 with specific prompts to generate a list of keywords. If the model fails to produce keywords meeting the criteria, an empty list is returned.

To validate the semantic alignment and expressive accuracy of the generated keywords, human annotators are engaged for manual evaluation. The acceptance criterion is set such that if the generated keywords achieve a matching rate of 80% or higher with their corresponding queries and document segments, the sample is considered acceptable. Based on this criterion, the overall acceptance rate of multi-scale keywords is 76.2%.

D. Reference Answer Generation

To serve as a benchmark for evaluating RAG systems, we generate high-quality reference answers for each filtered sample. The generation process similarly relies on prompt engineering, using deepseek-v3 to synthesize answers based on both the query and corresponding document segments.

Human annotators rate the generated answers on two dimensions (1–5 scale):

- **Fluency:** Assessing whether the response conforms to natural language usage, with clear grammatical structure and appropriate word choice;

- **Contextual Alignment:** Evaluating whether the answer fully leverages the provided document content, maintains strong semantic relevance to the query, and is free from deviation or fabricated information.

Only answers scoring 4 or higher on both dimensions were included in the final dataset, resulting in an acceptance rate of 89.4% at this stage.

E. Data Statistics

Through the three-stage filtering process described above, we construct a high-quality and well-structured evaluation dataset. The overall human screening acceptance rates are as follows:

- Query generation: 83.2%
- Multi-scale keywords: 76.2%
- Reference answer generation: 89.4%

After integrating these three stages, the final proportion of valid samples is 63.5%, indicating that challenges remain in the data generation and filtering process.

We further analyze the distribution of query types in the final evaluation set, as shown in Table III. The results indicate:

- Factual questions account for 47.8%, reflecting their widespread coverage in the documents and relatively low dependency on context, which makes them easier for models to generate;
- Tutorial questions have a lower proportion, partly due to the inherently smaller presence of instructional content in the source documents, and partly because instructional queries often require longer contextual reasoning and integration, resulting in lower pass rates during the generation and screening stages.

TABLE III: Proportion of query types

Query Type	Proportion (%)
Analytical	23.4
Factual	47.8
Comparative	11.7
Tutorial	9.6
Summary	7.5

V. EVALUATION METRICS

A. Chunking Evaluation

Current mainstream RAG evaluation frameworks generally overlook explicit assessment of the chunking stage. Most approaches either completely exclude performance metrics for this phase or infer chunking quality indirectly through downstream module performance changes, such as retrieval effectiveness or question-answering accuracy. While this implicit evaluation can partially reflect the impact of chunking strategies, it suffers from limitations including the lack of systematic measurement standards, weak interpretability of the evaluation process, and difficulty in accurately pinpointing the source of performance issues.

To address this, we propose an explicit chunking evaluation metric that analyzes the retrieved chunks within the top-n

results using both coarse-grained and fine-grained keywords. Specifically, we first filter the top-n chunks by retaining only those containing coarse-grained keywords, where n is set relatively large to mitigate the influence of retrieval performance on chunk evaluation. This is necessary because the retrieval mechanism may result in three scenarios: (i) candidate chunks appear in the middle positions of the top-n list; (ii) candidate chunks appear at the tail end of the top-n list; and (iii) candidate chunks are not retrieved at all. These scenarios can affect the selection of candidate chunks. Therefore, by employing a broad retrieval recall range combined with coarse-grained keyword filtering, we explicitly identify and retain candidate chunks suitable for evaluation, thereby reducing retrieval interference in chunking assessment.

Based on this, we designed two explicit metrics: Keyword Coverage Rate and Keyword Density.

Keyword Coverage Rate (KCR): A high-quality chunk should cover the core semantics relevant to the query. We introduce the Keyword Coverage Rate to measure the extent to which a text chunk retains the relevant keywords. This metric reflects how many target keywords are actually present within the chunk; a higher coverage rate indicates that the chunk is more likely to contain the information needed by the user query. The specific calculation is as follows:

Let the keyword set for the i-th sample be $K_i = \{k_{i1}, k_{i2}, \dots, k_{in}\}$. The set of keywords that actually appear in all candidate chunks for the i-th sample is denoted by $K'_i \subseteq K_i$. Then, the Keyword Coverage Rate for the i-th sample is defined as:

$$\text{KCR}(K_i) = \frac{|K'_i|}{|K_i|} \quad (1)$$

Keyword Density (KD): In addition to covering as many keywords as possible, a high-quality chunk should also maintain a high density of keywords to avoid sparsity of key information. We introduce Keyword Density to measure the frequency of target keyword occurrences within a text chunk, reflecting the focus of semantic content. The calculation process is as follows:

Let the keyword set for the i-th sample be $K_i = \{k_{i1}, k_{i2}, \dots, k_{in}\}$, $k_j \in K_i \cap C_i$ denote keywords that appear in the concatenated candidate chunks C_i for the i-th sample. Let f_j represent the frequency of keyword k_j in C_i , and $\text{len}(k_j)$ denote the length (in terms of word count) of keyword k_j . Let L be the total length (in words) of C_i . The Keyword Density for the i-th sample is then defined as:

$$\text{KD}(C_i, K_i) = \frac{\sum_{k_j \in K_i \cap C_i} \text{len}(k_j) \cdot f_j}{L} \quad (2)$$

Finally, we combine these two metrics to reflect the overall quality of chunking. The final score is calculated as:

$$\text{Score}(C, K) = 1 - e^{-\frac{\gamma}{N} \sum_{i=1}^N \text{KD}(C_i, K_i) \cdot \text{KCR}(K_i)} \quad (3)$$

where γ is a scaling factor set to 100.

Keyword Density focuses on measuring the sparsity of keywords within chunks, while Keyword Coverage Rate emphasizes the chunk's direct ability to capture target keywords, reflecting its semantic alignment with the query. Together, these two metrics evaluate chunk quality from the perspectives of structural coherence and semantic coverage, respectively. They effectively reveal the strengths and weaknesses of chunking strategies in terms of information density and semantic retention. Compared to traditional indirect evaluation methods that rely on downstream performance variations, these metrics offer advantages in quantifiability, interpretability, and reusability.

B. Retrieval and Reranking Evaluation

In the evaluation of the retrieval and reranking stages, we introduce a dual-metric system based on keywords to measure both the recall and precision of the retrieved results. Specifically, for each query, we retain the top k retrieved document chunks (top-k candidates), concatenate them into a final contextual input segment, and compute two core evaluation metrics based on fine-grained keywords.

Keyword Recall: This metric measures the proportion of target keywords successfully retrieved for each query. Specifically, it is defined as the ratio of the number of keywords retrieved in the contextual input segment to the total number of keywords for that sample, reflecting the system's ability in semantic coverage. Formally, the recall is calculated as:

$$\text{Recall} = \frac{1}{N} \sum_{i=1}^N \frac{TP_i}{Total_i} \quad (4)$$

Where TP_i denotes the number of coarse-grained and fine-grained keywords contained in the contextual input segment for the i-th sample, and $Total_i$ represents the total number of corresponding keywords for that sample.

Sample Accuracy: Defined as the proportion of samples for which the candidate text chunks fully cover all the target keywords. Formally, accuracy is calculated as:

$$\text{Accuracy} = \frac{Correctnessect}{Total} \quad (5)$$

where $Correctnessect$ denotes the number of samples in which all target keywords appear within the candidate text chunks, and $Total$ represents the total number of samples.

C. Generation Evaluation

To comprehensively assess the quality of generated responses, we employ a combination of automated and human evaluation metrics, encompassing classic methods in natural language generation as well as evaluation dimensions specific to RAG systems. Specifically, these include:

(1) **Bilingual Evaluation Understudy (BLEU)** [36]: This metric measures the n-gram overlap between the generated text and the reference answer, reflecting lexical-level similarity. BLEU is sensitive to phrase structure consistency and is well-suited for evaluating the accurate expression of fact-based questions. The specific calculation formula is as follows:

$$BLEU = BP \times \exp\left(\sum_{n=1}^N W_n \log(P_n)\right) \quad (6)$$

where the brevity penalty (BP) is defined as:

$$BP = \begin{cases} 1, & \text{if } c \geq r \\ e^{1-\frac{r}{c}}, & \text{if } c < r \end{cases} \quad (7)$$

Here, c and r denote the length of the candidate and reference answers, respectively. P_n represents the precision of n -grams computed by:

$$P_n = \frac{\sum_{g \in G_n} \min(\text{Count}_{Pred}(g), \text{Count}_{Ref}(g))}{\sum_{g \in G_n} \text{Count}_{Pred}(g)} \quad (8)$$

Where $W_n = \frac{1}{N}$, and in this study, N is set to 2. G_n denotes the set of n -grams in the candidate answer.

(2) **Recall-Oriented Understudy for Gisting Evaluation (ROUGE-L)** [37]: ROUGE-L focuses on measuring the longest common subsequence (LCS) between the generated text and the reference answer. It emphasizes the preservation of sentence structure and word order, making it well-suited for evaluating the quality of summarization or analytical responses. Let:

- $LCS(Pred, Ref)$ be the length of the longest common subsequence between the predicted and reference texts;
- $|Pred|$ be the token count of the predicted answer;
- $|Ref|$ be the token count of the reference answer.

Then the precision score P , recall score R , and ROUGE-L are computed as follows:

$$P = \frac{LCS(Pred, Ref)}{|Pred|}, \quad R = \frac{LCS(Pred, Ref)}{|Ref|} \quad (9)$$

$$ROUGE - L = \frac{(1 + \beta^2)PR}{P + \beta^2 R} \quad (10)$$

where β is set to a large constant to emphasize recall.

(3) **Faithfulness**: Measures whether the generated response accurately reflects the provided context, aiming to avoid hallucination phenomena.

(4) **Relevance**: Assesses the semantic consistency between the generated answer and the original query, evaluating whether the system correctly understands the user's question and provides an appropriate response. This metric primarily reflects the system's query comprehension and contextual modeling capabilities.

(5) **Correctness**: Evaluates the factual accuracy of the generated content, with particular emphasis on entities, numerical information, and causal reasoning.

To implement the evaluation of faithfulness, relevance, and correctness, we employ the built-in faithfulness evaluator of the llamaindex toolkit, which uses Deepseek-v3 as the underlying model.

VI. EXPERIMENTS

We select three representative RAG systems—Naive RAG, Advanced RAG, and Self RAG—as the subjects for our experimental evaluation.

A. Chunking Evaluation

For each sample in the evaluation set, we apply four chunking methods: (1) sliding windows, (2) cluster chunking, (3) Kamradt chunking, and (4) recursive chunking. First, the top 30 retrieved chunks are selected as the initial candidate set for chunk evaluation. These candidates are then filtered using coarse-grained keywords. Finally, keyword density and keyword coverage rate within the candidate chunks are calculated to assess chunking quality. The experimental results are summarized in Table IV.

The results in Table IV reveal significant performance differences among the various chunking methods. Recursive Chunking demonstrates the most consistent performance across most task types, achieves the highest overall scores, and exhibits strong keyword coverage and balanced information distribution. Cluster Chunking performs particularly well on Analytical and Fact-based tasks, indicating its suitability for handling structured or factual queries. Kamradt Chunking shows advantages in certain tasks such as Summarization but is less effective for Comparative tasks. In contrast, the Sliding Windows approach consistently scores the lowest across all task types, suggesting that its fixed-window segmentation strategy has notable limitations in preserving key information.

B. Retrieval and Re-ranking Evaluation

For each sample in the evaluation set, we input it into different RAG systems and select the top 5 retrieved and re-ranked chunks as candidates for evaluation. We then assess their recall and accuracy rates. The experimental results are summarized in Table V. The results indicate that Self RAG demonstrates superior retrieval and re-ranking performance across most task types, particularly excelling in Fact-based, Comparative, and Tutorial queries. Specifically, Self RAG achieves the highest recall and accuracy rates in these categories, highlighting its effectiveness in locating and filtering relevant content. For example, in Fact-based tasks, Self RAG attains a recall of 0.696 and an accuracy of 0.708, significantly outperforming the other systems.

C. Generation Evaluation

The quality of generated responses directly reflects the overall performance of RAG systems. Since different RAG systems employ varying processing mechanisms at each stage, these differences have a direct impact on the quality of the final generated answers. We evaluated each question in the assessment dataset by comparing the generated answers with the reference answers using five metrics: BLEU, ROUGE-L, faithfulness, relevance, and correctness. The experimental results are presented in Table VI.

For Analytical queries, Self RAG significantly outperforms other systems, achieving the highest scores in BLEU (0.492), ROUGE-L (0.587), and Faithfulness (0.649). It also leads substantially in Correctness (4.004) compared to Naive RAG (3.125), indicating Self RAG's superior capability in integrating structured and logical reasoning information for analysis tasks. In Comparative queries, Self RAG continues to

TABLE IV: Chunking performance of different methods

Method	Scores				
	Analytical	Factual	Comparative	Tutorial	Summary
Sliding windows [38]	0.395	0.418	0.331	0.338	0.373
Cluster chunking [39]	0.496	0.512	0.427	0.444	0.492
Kamradt_chunk [39]	0.484	0.536	0.366	0.447	0.504
Recursive_chunk [39]	0.505	0.510	0.443	0.456	0.493

TABLE V: Retrieval performance of different methods

Method	Analytical		Factual		Comparative		Tutorial		Summary	
	Recall	Accuracy	Recall	Accuracy	Recall	Accuracy	Recall	Accuracy	Recall	Accuracy
Naive RAG [1]	0.623	0.605	0.656	0.655	0.616	0.652	0.685	0.638	0.599	0.578
Advanced RAG [40]	0.656	0.629	0.668	0.668	0.638	0.594	0.684	0.640	0.595	0.582
Self RAG [17]	0.629	0.621	0.696	0.708	0.673	0.637	0.692	0.656	0.600	0.594

TABLE VI: Generation performance of different methods

Type	Method	BLEU	Rouge-L	Faithfulness Score	Relevance Score	Correctness Score
Analytical	Naive RAG	0.376	0.418	0.421	0.376	3.125
	Advanced RAG	0.439	0.444	0.589	0.394	3.388
	Self RAG	0.492	0.587	0.649	0.414	4.004
Comparative	Naive RAG	0.322	0.391	0.458	0.353	3.396
	Advanced RAG	0.397	0.394	0.472	0.367	3.590
	Self RAG	0.436	0.528	0.501	0.484	4.126
Factual	Naive RAG	0.410	0.423	0.461	0.395	3.867
	Advanced RAG	0.432	0.448	0.520	0.400	4.020
	Self RAG	0.439	0.549	0.603	0.439	4.312
Summary	Naive RAG	0.344	0.441	0.425	0.364	3.240
	Advanced RAG	0.312	0.416	0.431	0.342	3.350
	Self RAG	0.394	0.441	0.544	0.369	3.790
Tutorial	Naive RAG	0.329	0.414	0.467	0.323	3.381
	Advanced RAG	0.311	0.453	0.441	0.376	3.690
	Self RAG	0.422	0.521	0.578	0.410	4.270

demonstrate its advantage, particularly in semantic relevance (Relevance evance 0.484) and correctness (4.126), reflecting its stronger comparative reasoning and logical clarity. For Factual queries, Self RAG attains a high Faithfulness score of 0.603 and a Correctness score of 4.312, highlighting its robust generation performance and stronger information fidelity in fact-oriented question answering. In Summarization queries, Self RAG also achieves the best Faithfulness (0.544) and Correctness (3.79) scores. Although its ROUGE-L score is comparable to Naive RAG, Self RAG maintains an edge in content accuracy and summarization capability. For tutorial queries, Self RAG obtains the highest ROUGE-L (0.521), Faithfulness (0.578), and Correctness (4.27), demonstrating its strength in generating well-structured, clear, and effective step-by-step responses.

VII. CONCLUSION

In this paper, we propose UniEval-RAG, a unified evaluation framework that comprehensively covers four critical stages of RAG systems: chunking, retrieval, re-ranking, and answer generation. UniEval-RAG addresses the gaps in existing RAG evaluation methods by providing end-to-end process coverage and explicit metrics for the chunking phase. By introducing metrics such as keyword density, keyword coverage rate,

and a multi-scale keyword system, the framework enables holistic assessment of chunk quality, retrieval and re-ranking relevance, and answer generation accuracy. Leveraging an evaluation dataset encompassing five query types, our experiments demonstrate that UniEval-RAG offers strong interpretability and sensitivity, effectively identifying performance bottlenecks and facilitating fine-grained comparison and optimization across different RAG systems. This work contributes a generalizable evaluation tool to support the development of more reliable and efficient RAG systems.

REFERENCES

- [1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel *et al.*, “Retrieval-augmented generation for knowledge-intensive nlp tasks,” *Advances in neural information processing systems*, vol. 33, pp. 9459–9474, 2020.
- [2] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” *Advances in neural information processing systems*, vol. 30, 2017.
- [3] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever *et al.*, “Language models are unsupervised multitask learners,” *OpenAI blog*, vol. 1, no. 8, p. 9, 2019.
- [4] J. Wei, Y. Tay, R. Bommasani, C. Raffel, B. Zoph, S. Borgeaud, D. Yogatama, M. Bosma, D. Zhou, D. Metzler *et al.*, “Emergent abilities of large language models,” *arXiv preprint arXiv:2206.07682*, 2022.

- [5] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray *et al.*, “Training language models to follow instructions with human feedback,” *Advances in neural information processing systems*, vol. 35, pp. 27730–27744, 2022.
- [6] Y. Feng, H. Hu, X. Hou, S. Liu, S. Ying, S. Du, H. Hu, and Y. Gao, “Hyper-rag: Combating llm hallucinations using hypergraph-driven retrieval-augmented generation,” *arXiv preprint arXiv:2504.08758*, 2025.
- [7] G. Izacard, P. Lewis, M. Lomeli, L. Hosseini, F. Petroni, T. Schick, J. Dwivedi-Yu, A. Joulin, S. Riedel, and E. Grave, “Atlas: Few-shot learning with retrieval augmented language models,” *Journal of Machine Learning Research*, vol. 24, no. 251, pp. 1–43, 2023.
- [8] L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, Q. Chen, W. Peng, X. Feng, B. Qin *et al.*, “A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions,” *ACM Transactions on Information Systems*, vol. 43, no. 2, pp. 1–55, 2025.
- [9] J. Liu, R. Ding, L. Zhang, P. Xie, and F. Huang, “Cofe-rag: A comprehensive full-chain evaluation framework for retrieval-augmented generation with enhanced data diversity,” *arXiv preprint arXiv:2410.12248*, 2024.
- [10] A. Salemi and H. Zamani, “Evaluating retrieval quality in retrieval-augmented generation,” in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024, pp. 2395–2400.
- [11] J. Saad-Falcon, O. Khattab, C. Potts, and M. Zaharia, “Ares: An automated evaluation framework for retrieval-augmented generation systems,” *arXiv preprint arXiv:2311.09476*, 2023.
- [12] G. Xiong, Q. Jin, Z. Lu, and A. Zhang, “Benchmarking retrieval-augmented generation for medicine,” in *Findings of the Association for Computational Linguistics ACL 2024*, 2024, pp. 6233–6251.
- [13] T. Kwiatkowski, J. Palomaki, O. Redfield, M. Collins, A. Parikh, C. Alberti, D. Epstein, I. Polosukhin, J. Devlin, K. Lee *et al.*, “Natural questions: a benchmark for question answering research,” *Transactions of the Association for Computational Linguistics*, vol. 7, pp. 453–466, 2019.
- [14] M. Joshi, E. Choi, D. S. Weld, and L. Zettlemoyer, “Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension,” *arXiv preprint arXiv:1705.03551*, 2017.
- [15] W. Shen, M. Wang, Y. Wang, D. Chen, J. Yang, Y. Wan, and W. Lin, “Are we on the right way for assessing document retrieval-augmented generation?” *arXiv preprint arXiv:2508.03644*, 2025.
- [16] T. Lan, D. Cai, Y. Wang, H. Huang, and X.-L. Mao, “Copy is all you need,” 2023. [Online]. Available: <https://arxiv.org/abs/2307.06962>
- [17] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, “Self-RAG: Learning to retrieve, generate, and critique through self-reflection,” in *The Twelfth International Conference on Learning Representations*, 2024. [Online]. Available: <https://openreview.net/forum?id=hSyW5go0v8>
- [18] P. Sarthi, S. Abdullah, A. Tuli, S. Khanna, A. Goldie, and C. D. Manning, “Raptor: Recursive abstractive processing for tree-organized retrieval,” in *The Twelfth International Conference on Learning Representations*, 2024.
- [19] S.-Q. Yan, J.-C. Gu, Y. Zhu, and Z.-H. Ling, “Corrective retrieval augmented generation,” 2024.
- [20] X. He, Y. Tian, Y. Sun, N. Chawla, T. Laurent, Y. LeCun, X. Bresson, and B. Hooi, “G-retriever: Retrieval-augmented generation for textual graph understanding and question answering,” *Advances in Neural Information Processing Systems*, vol. 37, pp. 132876–132907, 2024.
- [21] J. Zhao, Z. Ji, Z. Fan, H. Wang, S. Niu, B. Tang, F. Xiong, and Z. Li, “Moc: Mixtures of text chunking learners for retrieval-augmented generation system,” *arXiv preprint arXiv:2503.09600*, 2025.
- [22] Y. Lyu, Z. Li, S. Niu, F. Xiong, B. Tang, W. Wang, H. Wu, H. Liu, T. Xu, and E. Chen, “Crud-rag: A comprehensive chinese benchmark for retrieval-augmented generation of large language models,” *ACM Transactions on Information Systems*, vol. 43, no. 2, pp. 1–32, 2025.
- [23] Y. Liu, L. Huang, S. Li, S. Chen, H. Zhou, F. Meng, J. Zhou, and X. Sun, “Recall: A benchmark for llms robustness against external counterfactual knowledge,” *arXiv preprint arXiv:2311.08147*, 2023.
- [24] S. Es, J. James, L. E. Anke, and S. Schockaert, “Ragas: Automated evaluation of retrieval augmented generation,” in *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, 2024, pp. 150–158.
- [25] S. Wang, E. Khramtsova, S. Zhuang, and G. Zuccon, “Feb4rag: Evaluating federated search in the context of retrieval augmented generation,” in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024, pp. 763–773.
- [26] H. Li, Y. Su, D. Cai, Y. Wang, and L. Liu, “A survey on retrieval-augmented text generation,” *arXiv preprint arXiv:2202.01110*, 2022.
- [27] C.-H. Tan, J.-C. Gu, C. Tao, Z.-H. Ling, C. Xu, H. Hu, X. Geng, and D. Jiang, “Tegtok: Augmenting text generation via task-specific and open-world knowledge,” *arXiv preprint arXiv:2203.08517*, 2022.
- [28] W. Lin, R. Billoshmi, B. Byrne, A. de Gispert, and G. Iglesias, “Li-rage: Late interaction retrieval augmented generation with explicit signals for open-domain table question answering,” in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, 2023, pp. 1557–1566.
- [29] M. Douze, A. Guzhva, C. Deng, J. Johnson, G. Szilvasy, P.-E. Mazaré, M. Lomeli, L. Hosseini, and H. Jégou, “The faiss library,” *arXiv preprint arXiv:2401.08281*, 2024.
- [30] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, H. Wang, and H. Wang, “Retrieval-augmented generation for large language models: A survey,” *arXiv preprint arXiv:2312.10997*, vol. 2, no. 1, 2023.
- [31] O. Khattab and M. Zaharia, “Colbert: Efficient and effective passage search via contextualized late interaction over bert,” in *Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval*, 2020, pp. 39–48.
- [32] J. Chen, H. Lin, X. Han, and L. Sun, “Benchmarking large language models in retrieval-augmented generation,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 16, 2024, pp. 17754–17762.
- [33] Y. Tang and Y. Yang, “Multihop-rag: Benchmarking retrieval-augmented generation for multi-hop queries,” *arXiv preprint arXiv:2401.15391*, 2024.
- [34] X. Bi, D. Chen, G. Chen, S. Chen, D. Dai, C. Deng, H. Ding, K. Dong, Q. Du, Z. Fu *et al.*, “Deepseek llm: Scaling open-source language models with longtermism,” *arXiv preprint arXiv:2401.02954*, 2024.
- [35] A. Liu, B. Feng, B. Xue, B. Wang, B. Wu, C. Lu, C. Zhao, C. Deng, C. Zhang, C. Ruan *et al.*, “Deepseek-v3 technical report,” *arXiv preprint arXiv:2412.19437*, 2024.
- [36] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, “Bleu: a method for automatic evaluation of machine translation,” in *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, 2002, pp. 311–318.
- [37] C.-Y. Lin, “Rouge: A package for automatic evaluation of summaries,” in *Text summarization branches out*, 2004, pp. 74–81.
- [38] H. Gong, Y. Shen, D. Yu, J. Chen, and D. Yu, “Recurrent chunking mechanisms for long-text machine reading comprehension,” *arXiv preprint arXiv:2005.08056*, 2020.
- [39] B. Smith and A. Troynikov, “Evaluating chunking strategies for retrieval,” Chroma, Tech. Rep., July 2024. [Online]. Available: <https://research.trychroma.com/evaluating-chunking>
- [40] I. Ilin, “Advanced rag techniques: an illustrated overview,” *Medium*, 2023.